08/07/2026

Modern Travel Platforms

Using MongoDB: Search, Vector Workloads, and Agentic AI



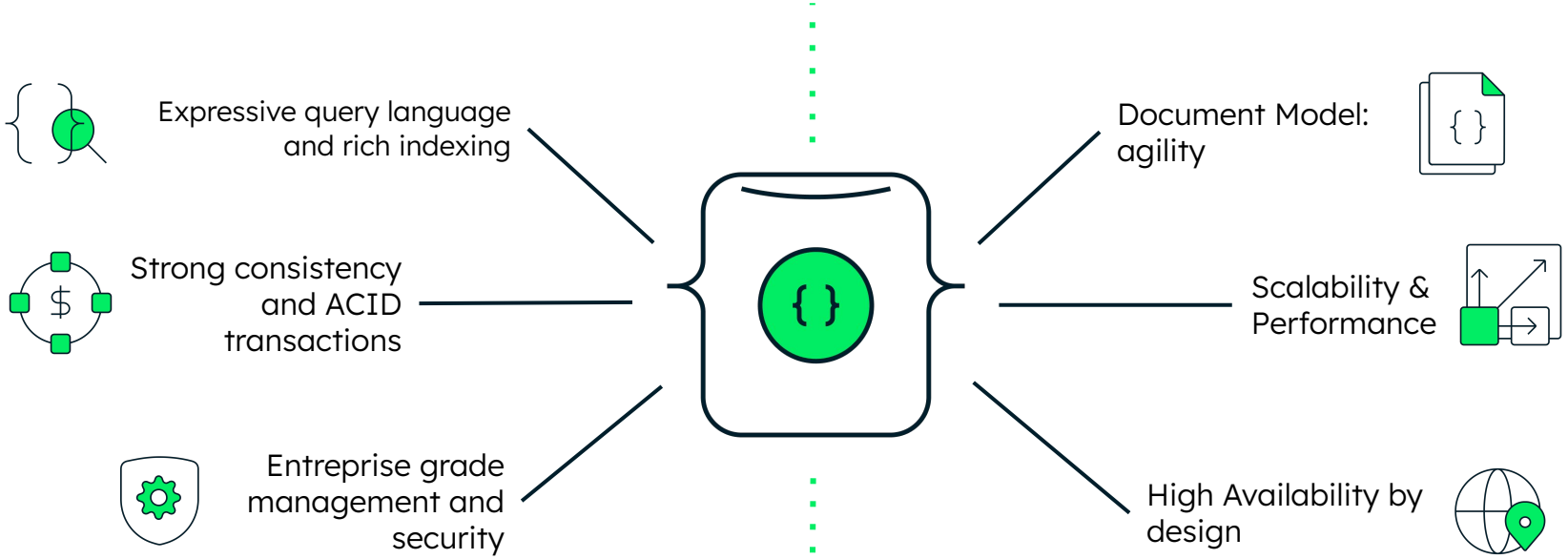
Thomas Scott

Senior Solutions Architect

{riviera dev}

MongoDB

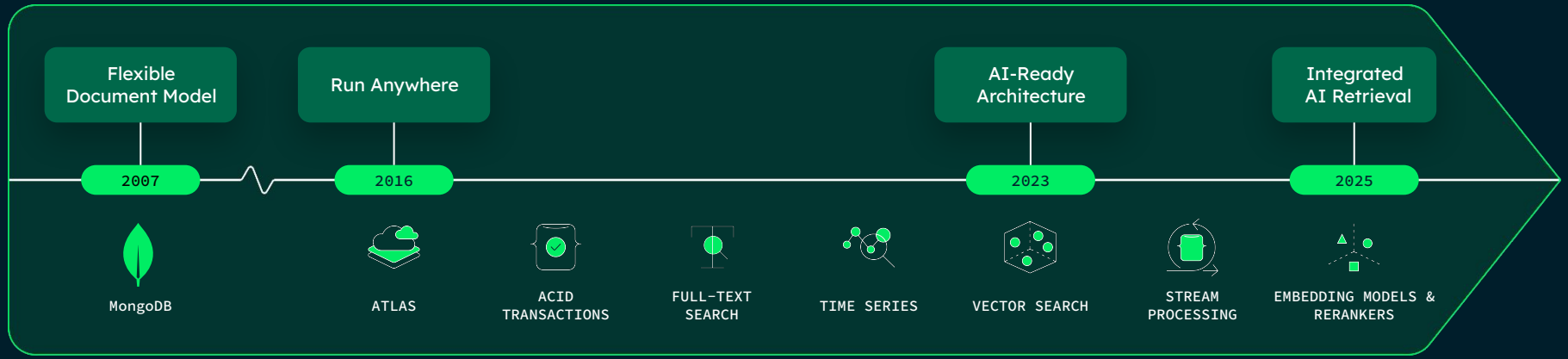
The best of both worlds



Traditional strengths of Relational

Strengths and Additional Innovations of NoSQL and Big Data

MongoDB was *built for change*, empowering businesses to innovate at the speed of the market



Seamless Scalability

End-to-End Security

Run Anywhere

What teams are trying to avoid, especially in modern travel platforms but not only....

The Complexity Tax

Multi-engine architectures create developer friction and operational risk.

01 **Separate operational database** for transactions.

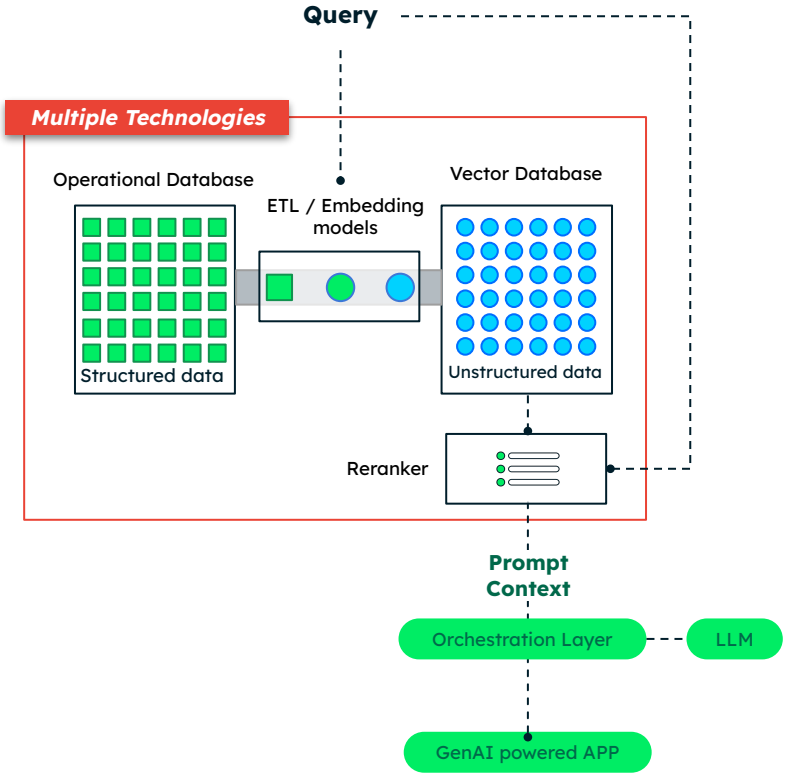
02 **Separate search engine** for keyword search and filtering.

03 **Separate vector store** for semantic retrieval and GenAI workloads.

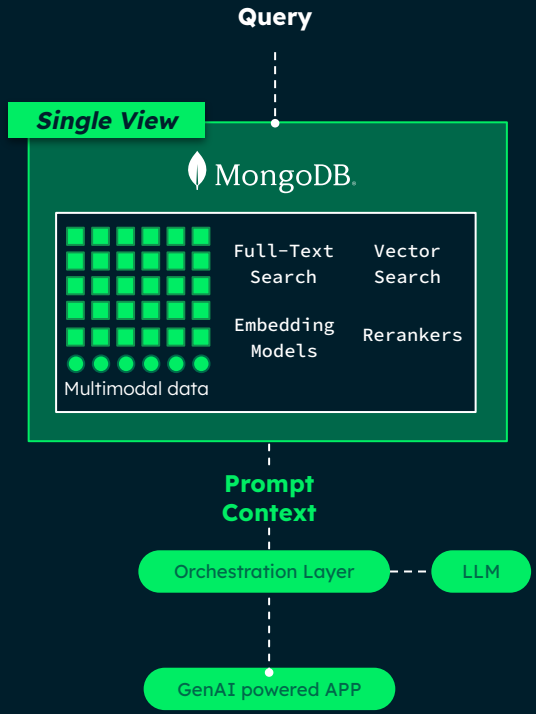
04 **ETL & synchronization pipelines** that introduce lag, cost, and failure points.

05 **Multiple operational overheads** including fragmented security, backups, and SLAs.

AI application data layer **without** MongoDB



AI application data layer **with** MongoDB



Five Pillars.
One Answer.
Why Us?

01 One Data Platform.

02 Built for AI.

03 Trustworthy, secure, governed by default.

04 Build once. Run anywhere.

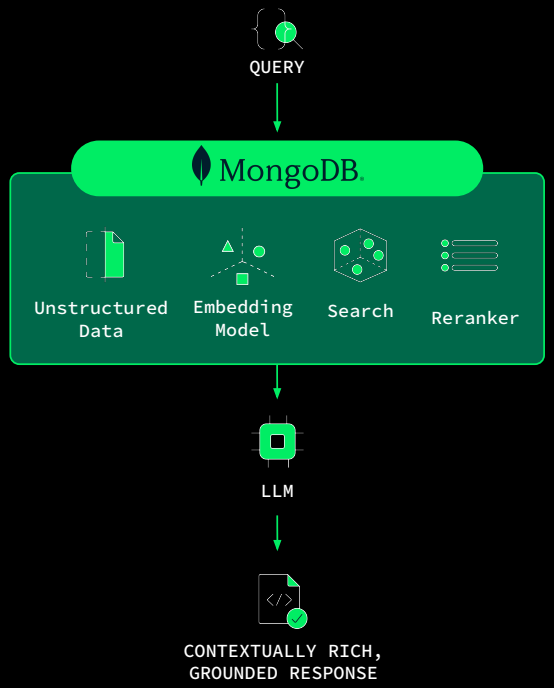
05 Control Context. Control Costs.

One Data Platform.

One API, any AI workload
Search, vector, embeddings, hybrid retrieval, time series, and streaming run on the same OLTP foundation

Eliminate the sync tax
Advanced retrieval capabilities where the operational data already lives eliminates costly syncing across systems

Workload Isolation without Data Isolation
Scalability and consistency required to run thousands of concurrent agents via Autoscaling and Dedicated Search Nodes



Built for AI.

JSON-native, the Lingua Franca of AI
Agents speak JSON natively, and so does MongoDB. Store context-rich, nested data in the format your AI already works in.

Flexible and adaptive schemas
Move as fast as your ideas. Schemas that adapt on the fly to any data type or structure without migrations and slowdowns.

Robust AI Partner Ecosystem
Native Voyage integrations with Amazon SageMaker, and Azure AI Foundry. Integrations with LLMs, and AI frameworks like LangChain, ground agents seamlessly in your business context

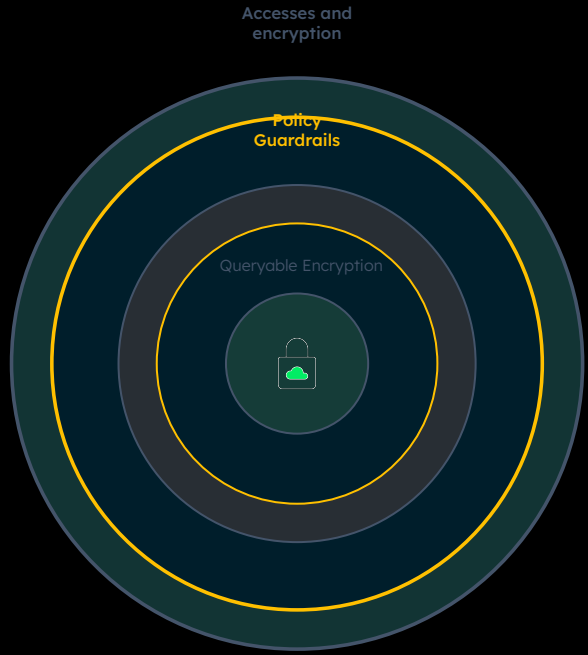


Trustworthy, secure, and governed by default.

Approved where it matters
Already trusted by the most regulated industries in the world. MongoDB's mature security model is cleared by corporate security at banks, hospitals, governments, and telcos.

Queryable Encryption
Search your sensitive data without ever exposing it. PII and PHI stay encrypted while being queried, not just at rest. Unique to MongoDB.

Guardrails agents inherit automatically
Agents can't bypass what the platform enforces. Field-level encryption, IAM, auditing, and policy enforcement are built in from the start.



Build once. Run anywhere.

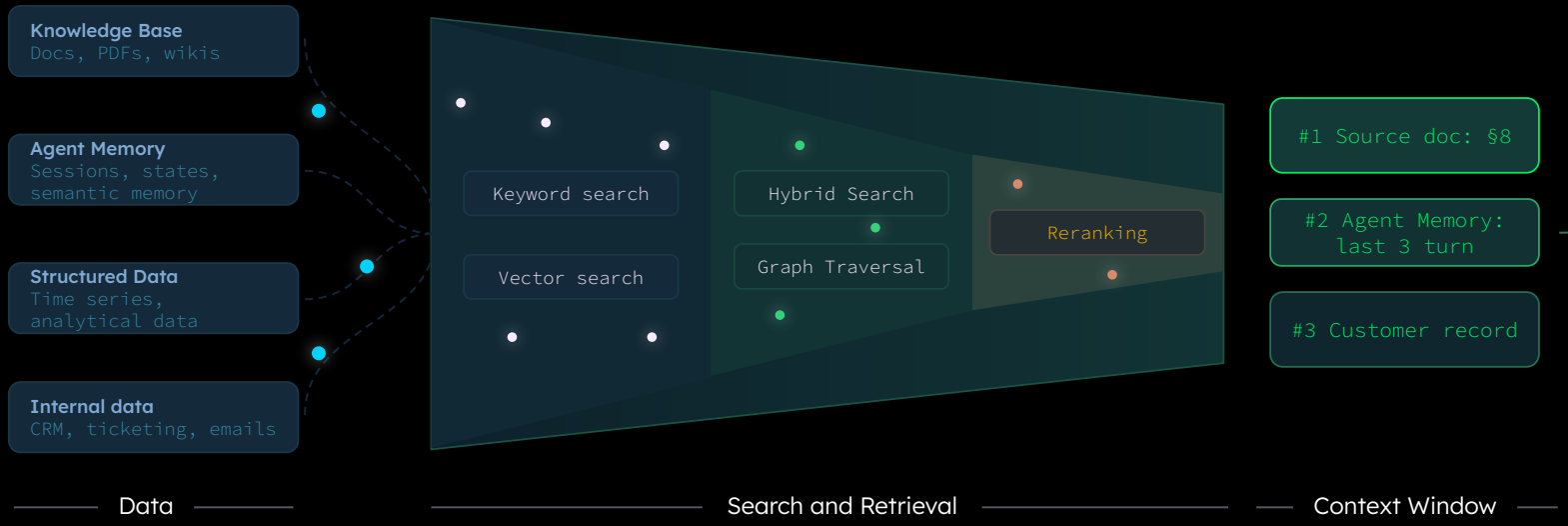
Multi-Cloud
Run your workloads wherever your data lives. MongoDB runs natively on AWS, Google Cloud, and Azure.

Local to production, seamlessly
From laptop to cloud and production without rewrites or config changes. Scale up when you're ready.

Source available, no vendor lock-in
Move between environments freely. Unlike Postgres flavors that lock you in with proprietary extensions, MongoDB keeps your code portable wherever you run it.



Enhance Context. Higher Accuracy. Lower Costs.



Search and AI are becoming increasingly fundamental across many application use cases

Application use cases

Search	AI-powered search	RAG	Agentic AI
Product search, document search, catalogs, image/ media search, etc.	Semantic search, recommendation engines, sentiment analysis, etc.	Chat with docs, natural language queries, chatbots, code and content generation, etc.	Multi-tool agents, customer support augmentation, intelligent workforce augmentation, etc.
Full-Text Search	Vector / Hybrid Search, Embedding Models, Rerankers	LLM + AI-Powered Search	LLM + Perception, planning, tools, memory

Search and AI implementations

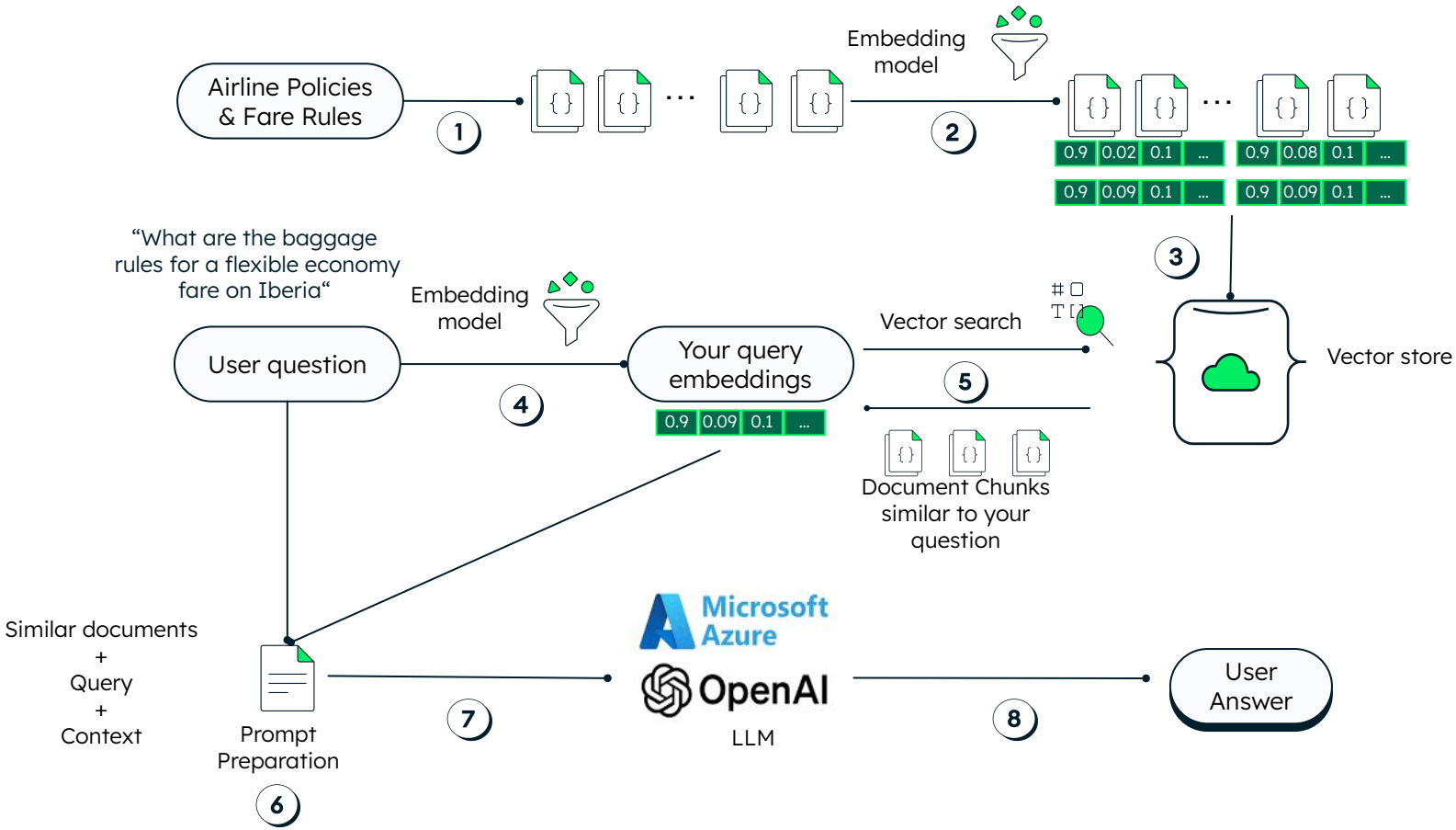
AI Use Cases

CAPABILITIES REQUIRED:

- ● ALWAYS
- ○ SOMETIMES
- ■ NOT NEEDED

	LLM	Embedding Model	Vector Search	Metadata Filtering	Hybrid Search	Reranker
RAG and Knowledge Q&A	●	●	●	○	●	●
Semantic Search		●	●	●	●	●
Recommendations	○	●	●	●	○	○
Agent Memory and Contextual AI	●	●	●	○	○	●
Security and Fraud Detection		●	●	●	○	

RAG Architecture





Announcing the LangChain + MongoDB Partnership: The AI Agent Stack That Runs On The Database You Already Trust

Build production AI agents on MongoDB Atlas — with vector search, persistent memory, natural-language querying, and end-to-end observability built in.

PARTNER POST 6 MIN READ MAR 31, 2026



“LangChain and MongoDB are turning MongoDB into a production-ready AI agent stack, unifying retrieval, memory, and data access in one platform - Harrison Chase”

Abstraction layers

Layer 4 - Harness (deepagents)
Planning, sub agents, a filesystem; you supply intent

Layer 3 - Agent (LangChain)
The agent loop prebuilt; you supply model, tools, and prompt



Uses

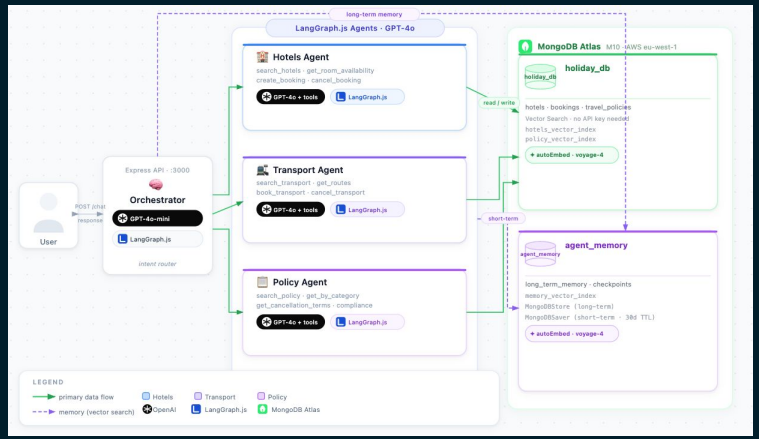
Layer 2 - Orchestration (LangGraph)
StateGraph, nodes, edges, checkpoints; you control the flow

Layer 1 - Building Blocks (LangChain-Core)
Models, messages, and tools; you control everything

Check out this sample repo I created

Enterprise-Ready AI Integration

Possible to do using MongoDB Enterprise Advanced. Build production AI agents with retrieval, memory, and action.



GitHub:
https://github.com/scott-thomas/multi-agent-holiday-assistant-mongodb

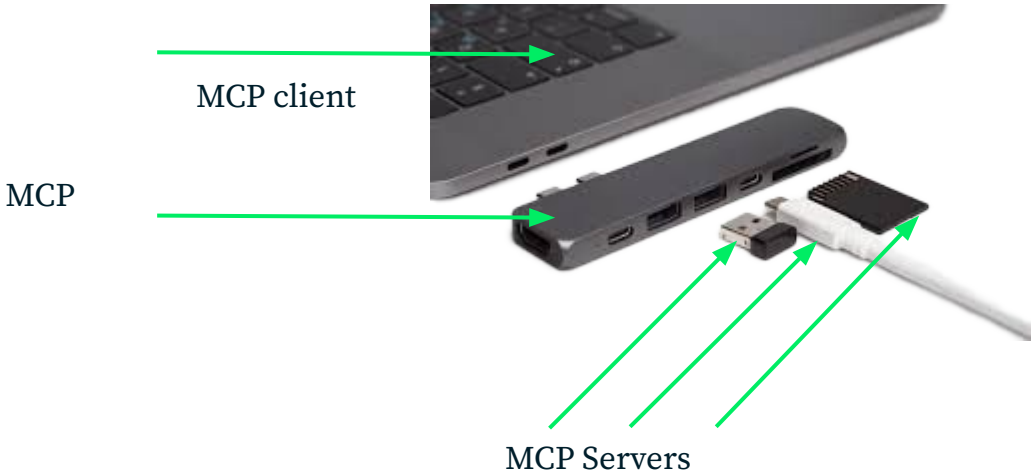
Scan to View Repository

multi-agent-holiday-assistant

LangChain × MongoDB

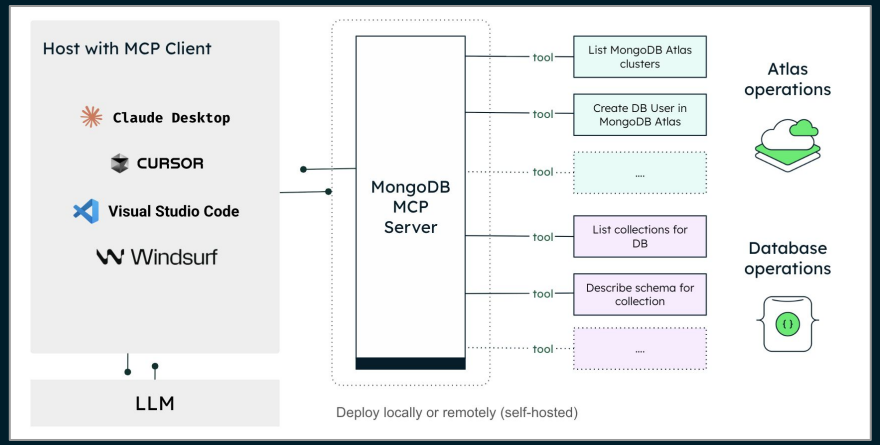
Demo Time

Model Context Protocol (MCP)



“The Model Context Protocol (MCP) is an **open protocol** that enables seamless integration **between LLM applications and external data sources and tools**. Whether you're building an AI-powered IDE, enhancing a chat interface, or creating custom AI workflows, MCP provides a standardized way to connect LLMs with the context they need.”

UNPARALLELED SIMPLICITY



Unique MCP Server

- Works with all MongoDB deployment options: Atlas, Community Edition, and Enterprise Advanced.

Flexible Configuration

- Robust configuration options to connect, secure, monitor, and customize deployments.

Comprehensive Tools

- Tools for databases and Atlas, including collection operations, Atlas project, and cluster management.

Scoped Authorization

- Secure operations through scoped, permissioned access using Atlas service accounts and database user.

Agentic Systems

Agentic systems **combine one or more AI agents with autonomous decision-making capabilities**, able to access and use resources to achieve defined objectives while adapting to environmental feedback.

MongoDB as a data foundation simplifies memory for AI agents

Modes of memory

Conversational, assistant, workflow

Tooling to enable business logic

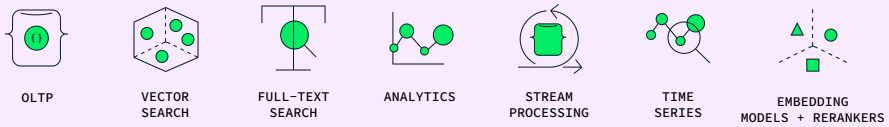
LangGraph

MCP through Tools

AutoGen

And more...

MongoDB.



Multi-Cloud Scale, Resilience, Performance, & Security
supporting short-term and long-term memory, and state persistence on multimodal data

End-to-end flow for agentic AI

01
User asks

02
Orchestrator runs RAG

03
Agent writes memory

04
Agent acts through MCP

05
Result persists

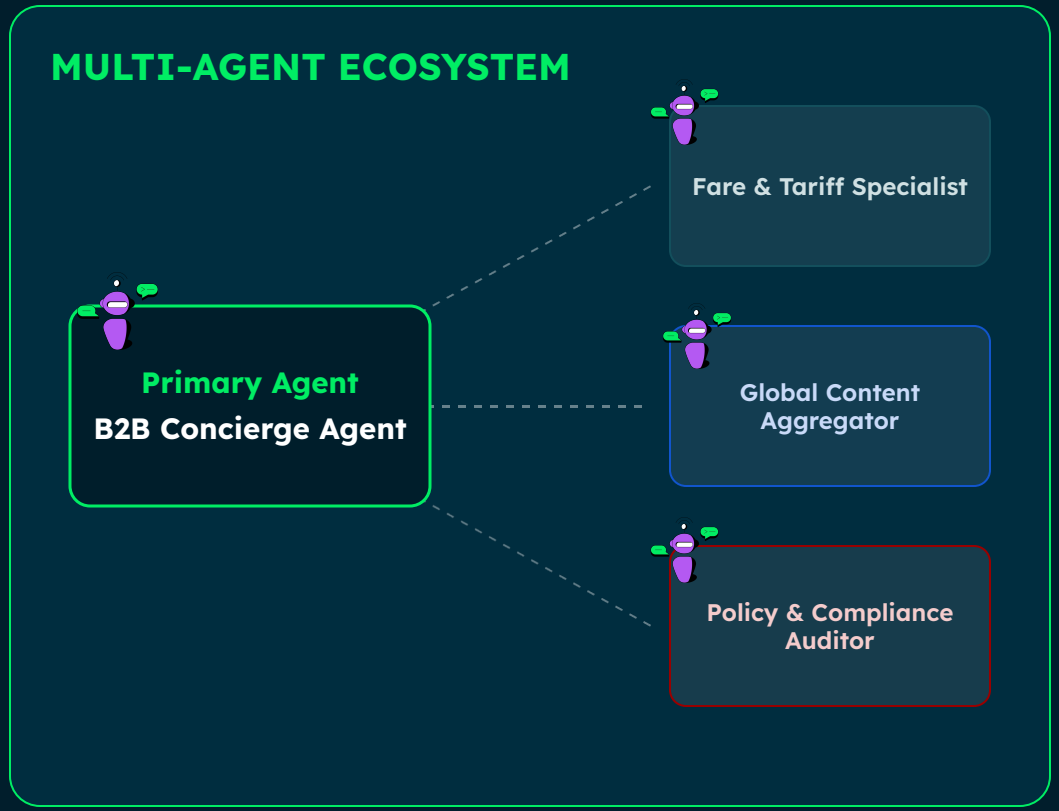
LangChain

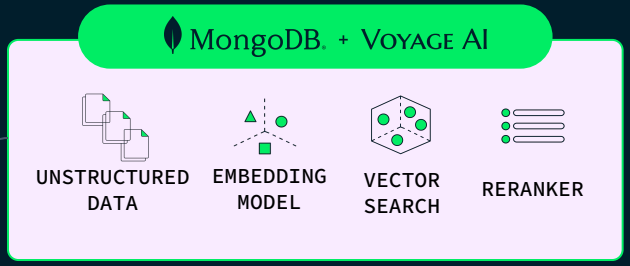
Voyage AI By MongoDB



MCP (Model Context Protocol)

Imagining Agentic AI Capabilities

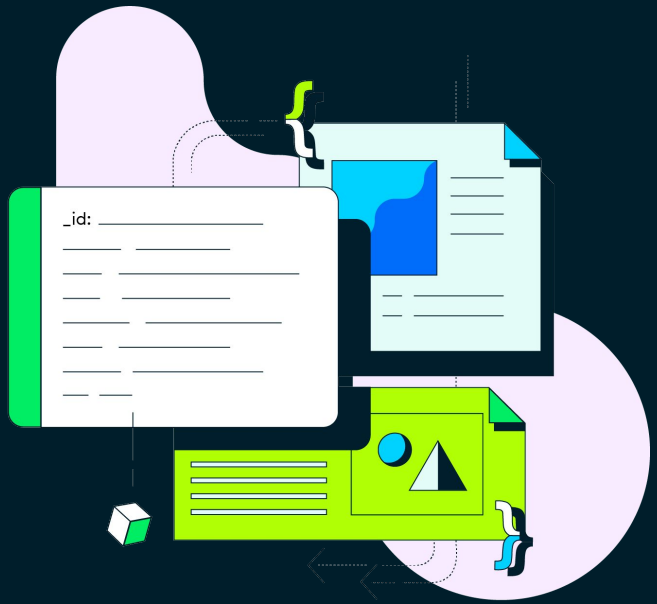




An **AI Agent** is a computational entity with:

- **Perception** through input
- **Action** via tool use
- **Cognition** through models
- **Memory** (long & short term)

New in MongoDB 8.2: Run MongoDB Search Anywhere



Build intelligent applications powered by lexical and semantic search over any type of data — now available on MongoDB Atlas and via public preview for use with Community Edition and Enterprise Server.

Search integrates advanced full-text search capabilities into applications.

Vector Search leverages semantic search and generative AI.

Hybrid search blends vector and keyword-search.

LangChain MongoDB

LangChain + MongoDB: Complete Agent Backend

The formal partnership announced in March 2026 positions MongoDB as a full-stack engine for persistent, stateful AI agents.

Memory & Retrieval

- Vector stores and retrievers.
- Chat history and semantic cache patterns.

State & Tooling

- LangGraph and LangSmith checkpointer support.
- Crash recovery and durable execution.
- Time-travel debugging.

Deployment

- Full support across MongoDB Atlas and EA
- Unified observability.

Three takeaways

01

Agents are a data problem

Context, memory, and actions matter as much as model quality.

02

Converge on one layer

Operational data, RAG, and memory fit better together than in separate stores.

03

Use standards

MCP plus LangChain/LangGraph gives a practical path to safe enterprise agents.

Thank you

Scan & Fill in to learn more



https://forms.gle/bbSrGBJtKAmkvkPE7